Chapter I: Introduction

1.0 Project Purpose:

In today's fast-paced, technology-driven world, emergency response carries extreme weight for the safety of the public and saving lives. But traditional ways are still being carried out by local groups that manage emergency response even though these old methods cannot meet the rising pressure of urgent situations. Today people expect faster communication, better coordination and reliable emergency services, especially in fire, medical emergency and natural disaster situations.

This is the current situation in Parañaque's Barangay Sun Valley. The emergency response is handled by BEMO, BDRRMO, POPSMO (Barangay Tanod) and the Brigada Onse Sun Valley Fire and Rescue Volunteer Group. They communicate and record incidents through radio, text message, chat apps and paper logs. These techniques do work, but they often create delays, miscommunication, incomplete records. Based on initial interviews with responders and dispatchers, when citizens request assistance, incident details must first be manually recorded before being relayed to responders. This manual contains many separate stages that must be done before the responders even arrive at the location of the incident which causes the arrival time to be delayed that sometimes range from 4-8 minutes. It has been observed by the U.S Fire Administration that the risk of injuries, fatalities and property damage is expanded when emergency response is slowed by even a small number of minutes.

This study intends to address these difficulties through the creation of a GIS-Based Emergency Dispatch and Coordination System with Real-Time Monitoring built for the exact requirements of Barangay Sun Valley. The main purpose of this project is to make emergency operations better through digital transformation by enhancing how involved personnel communicate and record incidents. The proposed system will be developed as a Progressive Web Application

(PWA) that functions on computers and mobile devices, which permits emergency personnel to oversee crises as they occur. The system will be utilized to provide a central platform, real-time incident monitoring and an SOS button feature to help citizens request emergency assistance. Through the implementation of the system, the study seeks to reduce response times, strengthen coordination among emergency response groups, and improve public safety within Barangay Sun Valley.

1.1 Project Description

Emergency response operations in Barangay Sun Valley are currently done manually through radio communication, text messages, chat applications, and paper-based records of BEMO (BDRRMO), POPSMO (Tanod) and other responders. Such methods are usually associated with a delay in coordination, inadequate documentation and slower response times. In some cases, the delay of emergency response may be about 4-8 minutes. Such a delay is critical in the case of fires and other life-threatening emergencies where a matter of minutes can make all the difference.

The proposed system will address current issues of the Barangay and will provide centralized dispatching, real time GIS and GPS tracking, digital incident management and mobile accessible SOS feature to improve coordination, reduce response delays, improve documentation and increase the overall efficiency of the barangay emergency response operations.

Key features include:

- Automated emergency call handling with auto-location pinning
- Real-time GIS mapping and GPS tracking of responders and units
- Digitalized transport scheduling to minimize the mistakes created in manual recording.
- Operational monitoring of the vehicles and equipment.

- Exportable digital reports for transparency and accessibility.
- Personnel management tools for the attendance, shift tracking and records.
- Mobile accessible SOS feature for citizens integrated within the Progressive Website Application.

The system is needed to reduce the operational delays and inefficiencies associated with the manual process and to improve coordination. This will also improve communication and real time decision making via a centralized digital platform.

1.2 Objectives of the Study

1.2.1 General Objective

The general objective of our project is to create and design a GIS based Emergency Dispatch and Coordination System with a Real time monitoring. By creating this system it will improve coordination, communication, incident management, and reduce operational response delays in Barangay Sun Valley emergency response team.

1.2.2 Specific Objectives

1. To develop a centralized dispatch system to manage emergency response incidents and coordinating responders.
2. To implement an accessible mobile SOS feature with real time incident reporting and location tracking.
3. To develop a GIS based monitoring system for a real time monitoring of the incidents, responder, and emergency system
4. To build a digital communication and documentation module for emergency alerts, transporting schedule, and incident records.

5. To implement a role based management system for the dispatcher, responder, and citizen with specific user access and personnel management features.

1.3 Significance of the Study

This study is significant in improving emergency dispatch and responder coordination at the barangay level through a centralized and technology-driven emergency management system.

- **Barangay Officials:** The system provides centralized monitoring and real-time incident tracking that can help reduce communication delays and improve coordination during emergencies.
- **Dispatchers (Central and Department):** The system simplifies emergency call handling, dispatching, and documentation processes, reducing time spent on manual processing and minimizing delays caused by fragmented communication.
- **Emergency Responders (BEMO and POPSMO):** Real-time alerts, GPS-based navigation, and incident updates improve response coordination and situational awareness during field operations.
- **Citizens:** The mobile-accessible SOS feature with automatic location sharing enables residents to report emergencies more quickly and accurately, helping reduce delays in emergency reporting.
- **Local Government Units (LGUs):** The system generates organized and exportable digital reports that support faster retrieval of incident records, performance evaluation, and data-driven planning for emergency preparedness.
- **IT Students, Researchers, and Future Developers:** This study serves as a reference for future GIS-based and Progressive Web Application (PWA) emergency systems and contributes to research and innovation in emergency management technologies.

1.4 Scope and Delimitation of the Study

1.4.1 Scope

This research centers on building a role-driven Progressive Web Application tailored for immediate emergency response within barangays. Designed for clarity, it supports unified command during crises by streamlining how alerts are assigned and managed from one hub. When someone sends an SOS, their position transmits automatically via GPS, cutting delays in locating help. Mapping unfolds through GIS tools, displaying both emergencies and available teams as they move across neighborhoods. Responders appear with status such as Available, En Route, On Scene and Transporting. Messages reach users through app alerts or SMS, ensuring updates arrive even without strong internet access. Each event logs digitally, forming structured reports that transfer easily into external files when needed

Most of the time, operation happens through an online setup so dispatchers, responders, and people can stay updated instantly. Even without a network, limited functions remain available because stored data lets users view key details and pending changes. Once connectivity returns, syncing resumes on its own. National emergency records won't link into this platform; features like forecast models or visual trend maps are also not included.

The main modules of the system include:

- Central and Department Dispatch Management
- Citizen SOS Feature with GPS Location Sharing
- GIS Mapping and Auto-Location Pinning
- Real-Time GPS Tracking with Unit Status Monitoring
- Notification System (SMS and App Alerts)
- Digital Documentation and Reporting System
- Field Responder Update Tools (status and media uploads)
- Personnel Management (attendance, shift tracking, training records)

1.4.2 Delimitation

While the system is to be built to enhance emergency response it has specific limitations to keep its focus narrow and not get too broad or confusing in terms of features and functionality.

- **Geographic Scope:** Only within Barangay Sun Valley will the implementation take place. Outside this area, events won't be managed by the platform. While designed specifically for local use, its reach stops at the boundary lines of that community.
- **Integration Limits:** The system will initially operate alone - no links to national emergency networks like 911, the PNP database, or BFP systems..
- **Connectivity Boundaries:** Most operations happen online, ensuring immediate updates. When the signal weakens, stored information remains accessible through local storage. Connectivity must exist to support live location monitoring. Alerts depend on network access for delivery. Functionality continues during brief outages due to buffered content.
- **Analytics Restrictions:** Basic reporting along with digital records will be handled by the system, though features like predictive modeling remain excluded. Advanced functions including AI-driven route planning are not part of its design. Disaster forecasting falls outside its capabilities entirely.
- **Hardware Boundaries:** This study focuses exclusively on software development. Items such as sirens, base stations, or specialized GPS equipment fall outside its boundaries. These external hardware elements may require distinct financial support. Implementation of such parts is not included here.
- **Device Compatibility:** Because the system operates as a Progressive Web Application, access to certain hardware functions can differ. Browser support influences what tools are available. Device type plays a role in functionality limits. Capabilities shift based on software environment. Not every feature works uniformly across platforms.

1.5 Review of Related Literature

When disaster strikes, knowing what is happening matters a lot. Built by Garcia et al., 2016, the Barangay Disaster Preparedness Monitoring Web Application System shows key details through visual displays, helping local government units see how ready individuals are. Because of this tool, local authorities can review community readiness fast - no long delays. Digital systems like these sharpen understanding of unfolding events, making responses quicker. Supporting this, Zhang et. al., 2023 showed that monitoring technologies - using sensors, network links, and analysis methods - help teams act faster, stay aligned, and grasp complex conditions under pressure.

Managing data accurately and keeping it in one place matters just as much. Found in work by Goh et al., 2019, a system built on community involvement uses database tools to gather details, generate reports, because it aids choices during emergencies. Results show typing up records digitally boosts how correct they are, accessibility, along with daily function efficiency. Similarly, Mercurio and Hernandez (2022) launched an Open Data and Geo-Based Information gathering illness cases, safety breaches, and calamity occurred. Improved record-keeping emerges through such design while also guiding response steps. Evidence across these projects confirms that such system structures help record emergency updates, vehicle plans, task lists methodically so people can retrieve them fast, reducing the possibility of having incomplete or overlooked information during critical events.

Involving the public improves how quickly geographic details are gathered. Introduced by Goodchild (2007), Volunteered Geographic Information (VGI) lets everyday people provide spatial observations using personal devices. Because of this, local events - like road dangers or emergencies - can be reported instantly via smartphone tools. Evidence from Klonner et al. (2016) shows these inputs strengthen responses during crises when combined with remote sensing outputs. Still,

inconsistencies in quality and difficulties linking datasets demand stronger verification steps within crisis response frameworks.

Today's emergency responses lean on shared digital tools to stay aligned. Without a "Common Operational Picture," coordination falters - Maharjan (2017) noted how groups like BDRRMO and POPSMO work apart, creating gaps. To fix this, one central system could link actions across teams, cutting repeat tasks or missed crises. Evidence backs it: De la Torre-Díez et al. (2022) showed mobile alert networks speed up reactions by sending live updates straight to field units, lowering mistakes caused by delay.

Using spatial data alongside GIS tools helps speed up emergency reactions. According to Luan et al., 2021, integrating GIS with decision-making frameworks aids in mapping safe zones as well as danger spots, assisting preparation efforts. In a similar way, research by Kurniawan et al., 2025, found route optimization becomes possible when live GPS feeds feed into geographic models - this also allows oversight of which units are ready. Instead of relying on guesswork, responders now follow insights drawn from current positioning and status updates.

Though often overlooked, surroundings shape how fast help arrives during crises. When gear is scarce, roads poor, or terrain tough, aid slows down - Smith and Brown showed this back in 2000. Because of such hurdles, spreading resources wisely becomes key. Especially where small communities manage with less, smart planning makes a difference.

Although less common, pairing handheld devices with central hubs can sharpen emergency responses. Developed by Machina in 2020, FEMACS - short for Fire Emergency Mobile Application and Central System - shows speed gains when field reports feed into a unified command structure.

Instead of rigid setups, adaptive frameworks allow smoother handling during crises. Efficiency rises when the system grows or shrinks based on demand.

Emerging tools like GIS, simulation frameworks, and smart online interfaces now shape how emergencies are managed. Built by Shih et al., one dispatch model combines geographic data with agent-based scenarios to adjust responses according to staff readiness, expertise levels, or willingness to take assignments. Results revealed faster deployment times - alongside a notable rise in lives saved, climbing as high as 15 percent - proving that decisions rooted in spatial evidence and real-time analytics can shift outcomes meaningfully.

Similarly, Yaneza et al., developed a web-based tool for handling emergencies more efficiently by combining geographic data, smart algorithms, and tools that aid decisions during crises. Starting from raw incident descriptions, it applies NLP along with TF-IDF methods to sort through text, spot key details, then suggest where help should go and how evacuations might unfold. When tested using ISO 25010 standards and measured through the lens of user acceptance theory, results showed strong performance - especially in consistency, ease of use, and approval from those operating it. This outcome underlines how smarter technologies can play a quiet but vital role when disasters strike.

Keerthika et al., meanwhile, introduced ResQlink - a tool built on web platforms meant for urgent incident logging, team coordination, and managing users. Built using up-to-date web frameworks, it uses location detection to track events precisely; at the same time, NLP helps sort emergencies by urgency. Security comes through access controls assigned by user roles. Their work shows that faster responses emerge when live updates meet protected information flow plus smart sorting of alerts. Though small in design, its impact lies where technology meets crisis handling.

Even though real-time tracking, geographic information tools, digital records, and smart coordination offer clear advantages, they are still underused in barangays. While many current research efforts look generally at disaster readiness or national frameworks, few target hyper-local solutions. To meet that unmet need, a new platform built on GIS-based Emergency Dispatch and Responder Coordination System that seeks to combine these capabilities within a role-specific Progressive Web App. This will enable unified oversight, live updates between responders, and smoother handling of critical information, ultimately strengthening how communities react during crises.

1.6 Review of Related Studies

These days, crisis management leans heavily on mobile devices, digital maps, real-time chat platforms, while tapping into virtual modeling - streamlining coordination under pressure. As systems evolve, responders gain quicker insights through current field reports, exact positioning data, alongside centralized control frameworks once emergencies unfold.

Though small in design, mobile tools reshape how fast help arrives when crises strike. Built on GPS signals, the app by de Guzman, de Guzman, and Ado (2014) feeds live updates straight to control hubs. Because locations transmit instantly, rescue teams adjust routes while en route. When danger hits, precise coordinates travel faster than spoken directions ever could. This shift cuts delays, linking people in distress directly to aid without detours. With every alert, clarity replaces guesswork in where help must go.

Just like earlier efforts, work by Nasir et al., 2022, introduced a mobile app interface built on Android meant to boost how quickly emergencies are reported. Instead of relying only on basic alerts, their design folded in personal data, types of crises, live positioning, along with images or

spoken notes when needed. Because it bundled more tools into one place, help arrived faster - communication became clearer, less delayed. What stood out was how much smoother things ran between the person pressing the alert and those coming to assist, showing just how useful smartphones can be during urgent moments.

Looking at mobile tech in disasters, Aisyah et al. (2026) reviewed multiple studies and discovered common use of GPS, alerts, along with messaging functions during planning, crisis, and rebuilding stages. Despite widespread integration, problems emerged - adoption remained weak, interfaces proved difficult, interaction dropped over time. Effectiveness is clear; still, designs need greater focus on access and real-user needs if uptake is to improve. Their work ends by underscoring function over feature overload.

Though mobile evacuation alerts can speed up reactions during crises, their success often depends on how they are delivered. When warnings come too frequently, people start ignoring them - like tuning out a constant beep. Design matters just as much as speed; getting it wrong weakens the entire system. A steady rhythm of urgent yet respectful messaging keeps attention without wearing it down.

Despite its technical core, Geographic Information Systems play a key role in real-time crisis handling. According to Aporto et al., 2026, mobile tools powered by GIS spot danger zones more accurately through incident pattern tracking. Because of this ability, teams can assign personnel and supplies where they're most needed. Spatial insights turn raw location data into actionable strategies during disasters - shaping how agencies prepare and react on the ground.

Though focused on modern tools, Dong (2024) pointed out how combining GIS with GPS plus mobile computing sharpens live tracking of locations. Because systems link together, routing

vehicles becomes more precise during urgent situations. When incident data appear instantly on maps, coordination shifts noticeably toward faster decisions. Since digital positioning feeds directly into field operations, teams find affected sites without delays. As a result, those managing emergencies gain clearer views of where help is needed most. Rather than relying on estimates, responders act using current geographic details shared across devices.

Similarly, Utami and Ramdani (2021) built GEMAR - a digital platform based on geographic information science meant for handling crisis alerts and sending out medical transport. One piece handles urgent reports by capturing location details through mapping tools. Meanwhile, another part calculates efficient paths for ambulances while pinpointing nearby treatment centers. Testing revealed strong performance across both ease of use and functional impact, highlighting how spatial technology can support rapid aid delivery when emergencies occur.

Not far from conventional approaches, Osebor et al. (2017) introduced a method using SMS to send emergency location alerts, aiming to speed up responses where network access is weak. Because older systems struggle with timing and precision, this work focused on fixing mismatched coordinates during crises. Despite basic tools, their design stayed consistent under pressure, showing usefulness in real situations. Though minimal in tech demands, it delivered steady performance when connectivity gaps exist.

Though common, standard methods often fail under pressure when seconds count. Dispatch decisions tied to place and moment require tools built for change, not fixed rules. A 2020 analysis by Mukhopadhyay and team framed emergencies as puzzles shaped by space, time, weather. When vehicles, personnel, and traffic shift unpredictably, old models fall short. Because of this mismatch, help sometimes arrives too late. Instead of static plans, fluid solutions adapt as situations unfold.

Surprisingly, Usanov et al., in 2019 turned to random-based simulations when analyzing how ambulances get sent out. Instead of just sending the closest one, their method considered unpredictable road delays - this shift improved outcomes noticeably. Because traffic patterns keep changing, along with where emergencies pop up, rigid rules fall short. Efficiency jumped once these moving parts were built into forecasts. Clearly, guessing what might happen beats assuming everything stays fixed.

One way to look at it - Green, Kolesar, and Whitt (2013) pointed out how math-based tools might sharpen emergency system performance. Instead of guessing, their research showed precise placement of ambulances, staff, and gear could shorten delays while expanding access during peak pressure moments. What stands out, though, is their reminder: real-world unpredictability must shape how these plans are built.

Across fields, tools using geographic data help shape how communities prepare for crises. Work by Muller and Zhang (2023) reveals such methods pinpoint vulnerable regions through mapping techniques. Because of these insights, emergency teams adjust where they place supplies and personnel. Visualization of danger zones enables planners to anticipate challenges before events unfold.

Johnson and Lee (2022) explored how information systems shape decision-making speed - revealing those handling vast, live data tend to sharpen accuracy while smoothing team alignment. When seconds count, such as during crises, their findings highlight a steady stream of updates as vital; even brief interruptions risk serious outcomes.

Even with progress made, much of today's research centers on standalone tools - mobile apps, for instance, or GIS platforms, or algorithmic planning methods. Integration remains rare

when it comes to merging live messaging, geographic visualization, unified command routing, team coordination, and electronic recordkeeping within one system. Many solutions are built more for cities or broad deployments instead of small community units like barangays.

To address this gap, the current project introduces a GIS-based tool for emergency dispatch and team coordination. Built on mobile connectivity and digital maps, it combines live location updates with a central command interface inside a web platform that works like an app. When information moves steadily, local groups become more ready. Readiness grows in small neighborhoods because updates keep coming. As details circulate without pause, barangay-level planning improves. With fresh inputs arriving often, collective readiness rises. Because data flows do not stop, community units adapt faster. Steady streams of facts help people prepare before crises hit.

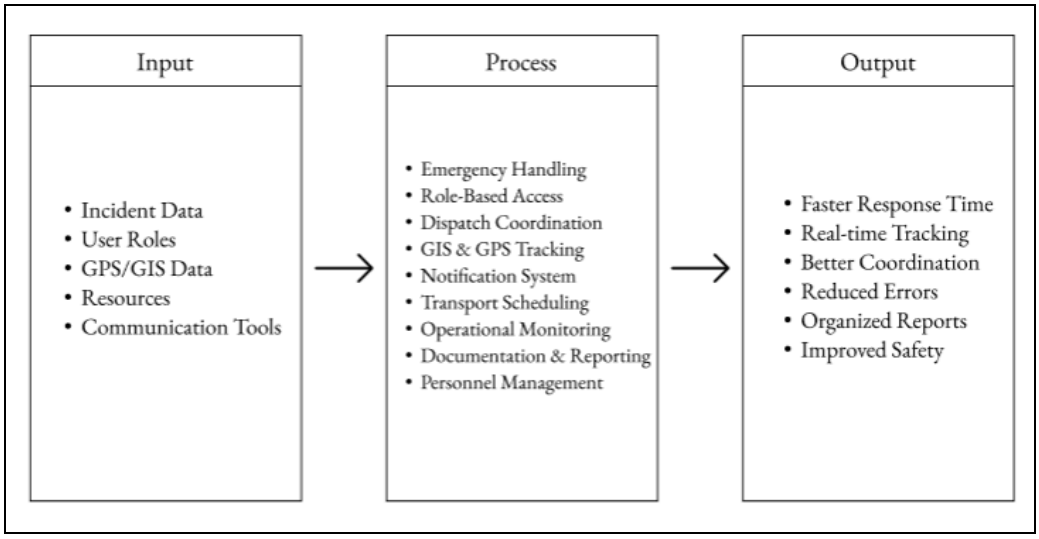
1.7 Technical Background

The system will be developed using a combination of web technologies, GIS mapping, GPS tracking, and Progressive Web Application (PWA) architecture. It will rely on:

- Database management systems for storing patient records, incident reports, and personnel data
- Notification services (SMS/App) for rapid communication
- GIS tools for mapping and incident visualization
- GPS tracking for real-time monitoring of responders
- PWA technologies (service workers, offline capabilities, and responsive design) to enable cross-device accessibility and app-like user experience

1.7.1 Conceptual Framework

The study is anchored on an **Input–Process–Output (IPO) model**, illustrating how the proposed project transforms emergency-related data into efficient and coordinated response outcomes.



1. Input:

- Emergency incident data (SOS requests, reports, incident types)
- User roles (citizens, dispatchers, responders)
- GPS and GIS data (location coordinates, mapping data)
- Available resources (vehicles, equipment, personnel)
- Communication tools (SMS/App notifications)

2. Process:

a) Emergency Request Handling

- Receiving SOS alerts from citizens
- Automatic location detection and pinning

b) Role-Based System Access

- Citizens: Send SOS requests

- Dispatchers: Monitor and manage incidents
- Responders: Receive alerts and update status

c) Dispatch Coordination
   - Assign nearest available responders
   - Coordinate between central and department dispatchers

d) GIS Mapping and GPS Tracking
   - Real-time incident visualization
   - Live tracking of responders and vehicles

e) Notification System
   - Send alerts and updates via SMS/App

f) Transport Scheduling
   - Manage patient transfers efficiently

g) Operational Monitoring
   - Track equipment and vehicle readiness
   - Update responder status

h) Documentation and Reporting
   - Automated report generation
   - Data storage and retrieval

i) Personnel Management
   - Attendance tracking
   - Shift scheduling
   - Training records

3. Output:

- Faster emergency response times
- Accurate real-time incident location tracking
- Improved coordination among dispatchers and responders
- Reduced manual documentation errors
- Centralized and organized reporting system
- Enhanced public safety and disaster preparedness

1.8 Definition of Terms

BEMO (BDRRMO): The Barangay Disaster Risk Reduction and Management Office responsible for planning, coordinating, and implementing disaster risk reduction and emergency response activities at the barangay level.

Dispatcher: An authorized personnel responsible for receiving emergency reports, assessing incidents, and coordinating the deployment of responders.

Geocoding: A process that converts addresses or place names into geographic coordinates (latitude and longitude) for use in digital maps and location tracking systems.

Geographic Information System (GIS): A system used for capturing, storing, managing, and analyzing spatial or location-based data to support mapping and decision-making.

Geolocation: A technology that identifies and determines the real-time geographic location of a device or user using GPS, internet connection, or mobile networks.

Global Positioning System (GPS): A satellite-based navigation system that provides real-time location tracking and positioning information.

Incident Monitoring: The real-time tracking, updating, and documentation of emergency events from reporting to resolution.

Operational Checklist: A digital tool used to monitor the readiness and status of emergency equipment, vehicles, and resources to ensure operational preparedness.

POPSMO (Tanod): Barangay peace and order personnel responsible for maintaining local security, assisting in emergency response, and supporting public safety operations.

Progressive Web Application (PWA): A web-based application that functions like a mobile app, allowing users to access the system on computers and mobile devices with features such as offline access, responsiveness, and real-time updates.

Transport Scheduling: The systematic assignment, coordination, and monitoring of patient or victim transport operations to ensure timely and efficient movement.

Chapter II. Methodology

2.1 Design of Software , System, Product or Process

2.1.1 Software Architecture

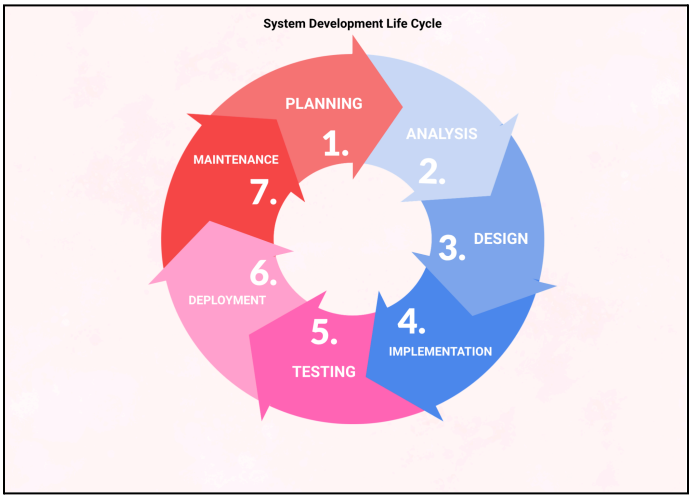
The software design is based on Progressive Web Application (PWA) that supports emergency operations at the barangay level through structured layers and user-specific access. While citizens participate directly, dispatch teams from BEMO or POPSMO manage alerts and assignments. Hospitals join alongside local police posts, each interacting according to predefined responsibilities. Role distinctions guide how individuals engage during crisis response. Communication flows smoothly because permissions align with operational duties. System structure ensures clarity when coordinating urgent actions. Citizens send out SOS requests. Dispatchers deal with incidents and assign responders. The responders, hospitals and police units are updated and take action as needed.

The system flow starts with an SOS request sent by a citizen through the mobile-accessible PWA which automatically gets the location of the citizen using GPS. The Central Dispatcher takes the request, evaluates the incident, and communicates with the appropriate department. The Dispatch Coordination module allows the dispatcher to assign the nearest available responders based on GIS mapping and real-time tracking. Responders get alerts via SMS or in-app notification, navigate to the location and update their status when on duty. It constantly tracks the progress of the incident and can notify other units such as hospitals or police depending on the nature of the emergency.

There will be several modules in the system that will work together to support the functions. The Incident Management module will be used for S.O.S. requests and reporting. Incidents and responders will be displayed in real time on a map using GIS mapping and GPS tracking. Communication will stay clear across teams through the Notification module while patient moves

will get organized by the Transport Scheduling part. Checklists for daily operations will appear alongside tools handling staff details, gear status, and supply levels. Frontend structure will take shape using React combined with Bootstrap styling elements. Node.js will form the server base, will work with Express.js, and will link directly into a MongoDB data store. Live updates will come alive by pulling in outside tools - Google Maps API will guide location work, GPS feeds will provide motion data, Firebase will handle messages, and Twilio will route alert texts.

2.1.2 Software Development Life Cycle



The Agile model allowed us to deliver in stages and accommodate change as it came. The only way we could work was indeed not building everything at once. This means that we can always iterate over every single feature and just work off one repository. Real-time location tracking through GPS is a part of complex features that simplified the processing of such things. This also worked as a step to ensure that the system was stable and effective prior to being used on such a scale. A major advantage of our agile process was gathering feedback from emergency responders to enhance app design. Instead, the system offered them key features they required. The initial training covered the most important features, such as the SOS button, while the remaining tools were scheduled for later releases.

1. Planning

In the planning phase, the team has specifically identified goals and features of a GIS Emergency Dispatch & Responder Coordination System. The operations of Barangay Sun Valley, the smallest administrative division in the Philippines, served as their study and found that a community-based system is the best way to improve emergency dispatching, incident tracking, responder coordination, and reporting. It also defined the project scope and what resources and technical specifications were needed to go in. With GIS mapping, GPS tracking, and SOS communication, we created a way to manage emergency responses and exchange real-time updates between users and responders. In addition, they pitched and conducted a feasibility study with resource allocation for the system to be deployed within the barangay's operational state.

2. Analysis

In the analysis stage, the team concluded that there was no centralized platform that unified the four key components within the current emergency coordination system: the citizen, dispatcher, administrator, and emergency responders. This leads to inefficiencies in decision-making and coordination. Poor communication of the various emergency units was blamed on reliance upon separate chat apps with no clear information. However, real-time tracking allows for effective task assignment and minimizes the response time. As records on paper were easily forgotten and transport schedules faded from memory, equipment was also frequently lost. These difficulties showed why a single platform is needed to combine all emergency response tasks.

3. Design

In the design phase, the team built the system framework from which cloud-based databases and an accessible interface would be accessed during emergency operations. Clean and simple UI

and UX designing with tools such as Figma. The major focus was GIS, where mapping layers were arranged to help the responders easily understand location information.

4. Implementation

In the implementation phase, it is built as a progressive web application (PWA) so that the solution can be accessed from desktop and mobile platforms. The citizens from Parañaque's Barangay Sun Valley will learn how to use features like SOS alerts, GPS tracking, incident reporting, and responder management with real-time dispatch. Sensitive information can be protected using user authentication and access control. A centralized database system for incidents and operations will be integrated with a GIS-based mapper module to visualize incidents & responders in real time. It also has features for scheduling, managing personnel, logistics, equipment monitoring, reporting, and real-time communication with dispatchers.

5. Testing

In the test phase, before launching, we'll test the system to check its speed, accuracy, and user friendly. We'll also have actual dispatchers and emergency responders test it out, like a process called user acceptance testing to make sure it meets their exact needs. It is used to confirm the system is operationally ready while also finding possible technical issues.

In addition to testing the SOS function and dispatch feature for stable emergency communication, a field trial in Barangay Sun Valley will verify the accuracy of GPS tracking and incident mapping. Simulating heavy traffic will also be used to identify potential bottlenecks in the database, user authentication, and reporting features. The test results to optimally design the interface system before implementation will also be used in testing.

6. Deployment

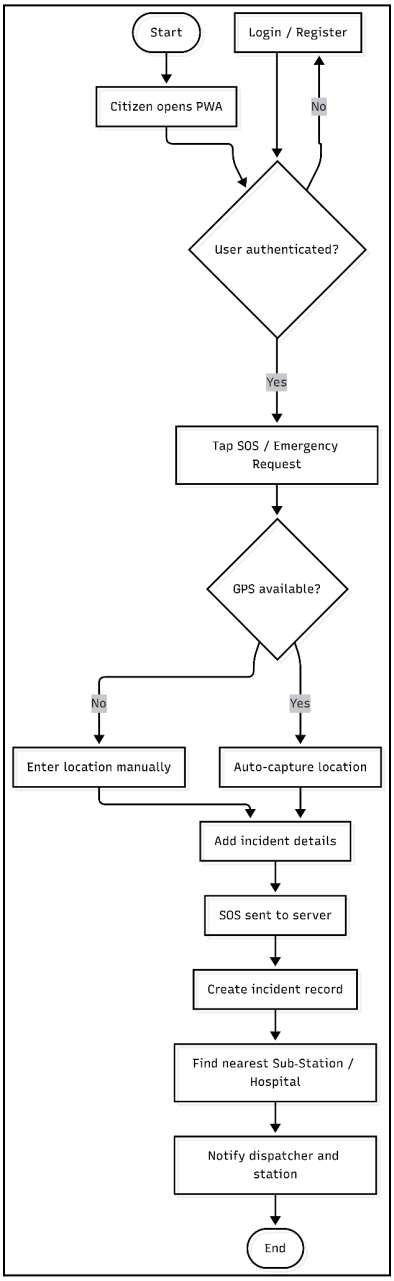
In the deployment phase, the system can be accessible by authorized users on phones or computers through a progressive web app (PWA). The system also will need configuration for dispatch coordination, GPS tracking capabilities, and incident reporting and responder monitoring. It shall implement access control by creating user accounts. Orientation sessions, along with some basic training, will also provide an overview of the native interface and functionality to set up users. Performance and areas needing technical assistance or adjustment will be evaluated in part through initial monitoring.

7. Maintenance

In the maintenance phase, team will monitor the platform, use real feedback from real users to fix bugs quickly. And regular updates will keep your data safe and make sure the app runs perfectly on any devices. This constant support allows the system to be ready for any changes in successful integrations, such as making improvements around offering map and reporting modules.

2.1.3 System Flowchart

Citizen Emergency Request Flow



Start → Citizen opens PWA

- The process starts when a user accesses the Progressive Web App on their device.

Login / Register → User authenticated

- -If the user is not authenticated, they must log in or register first.
- -If yes, the system proceeds.

Tap SOS / Emergency Request

- The citizen presses the SOS button to send an alert in case of an emergency.

GPS available?

- If GPS is available, the app automatically captures the user's location.
- If GPS is not available, the user must manually enter their location.

Add incident details

- The citizen provides further details about the emergency (e.g. type of incident, description).

SOS sent to server → Create incident record

- The app sends the request to the backend server.
- The server logs the incident in its database.

Find nearest Sub-Station / Hospital

- The system determines the closest emergency station or hospital to the incident area.

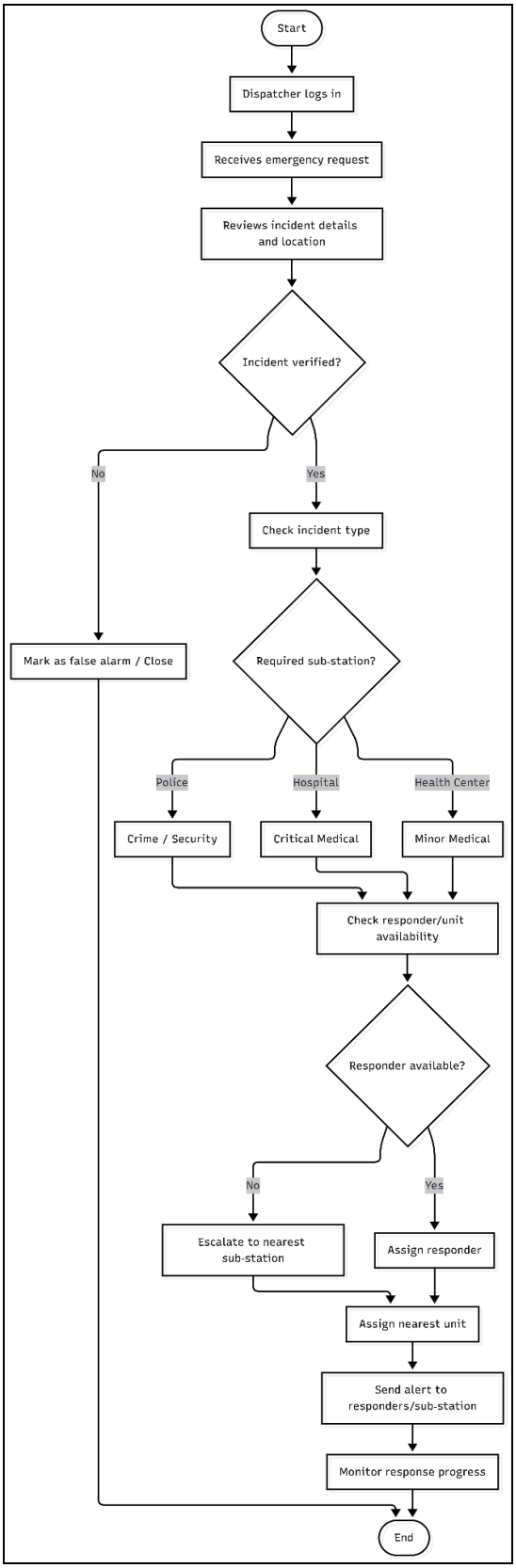
Notify dispatcher and station

- The dispatcher and relevant station are alerted with the incident details and location.

End

- The workflow completes once the notification is successfully sent.

Dispatcher Incident Handling Flow



Start → Dispatcher opens system dashboard

- The workflow begins when the dispatcher accesses the emergency management system and logs into the platform.

Dispatcher logs in

- The dispatcher enters valid credentials to gain access to the emergency response dashboard.

Receives emergency request

- The system receives an incoming SOS or emergency request submitted by a citizen through the PWA.

Reviews incident details and location

- The dispatcher checks the emergency information, including the incident description and GPS/location details provided by the citizen.

Incident verified?

The dispatcher determines whether the emergency request is valid or a possible false alarm.

- If no, the request is marked as a false alarm or closed.
- If yes, the process continues.

Check incident type

- The dispatcher identifies the category of the emergency incident to determine which responder or agency should handle it.

Required sub-station?

Based on the incident type, the system routes the request to the appropriate service:

- Police - for crime or security incidents
- Hospital - for critical medical emergencies
- Health Center - for minor medical concerns

Check responder/unit availability

- The dispatcher verifies whether responders or emergency units are currently available to handle the incident.

Responder available?

- If responders are available, the dispatcher assigns a responder to the incident.
- If no responders are available, the request is escalated to the nearest available sub-station.

Assign nearest unit

- The system determines and assigns the closest available responder or emergency unit to reduce response time.

Send alert to responders/sub-station

- Notifications and incident details are sent to the assigned responders and relevant emergency station.

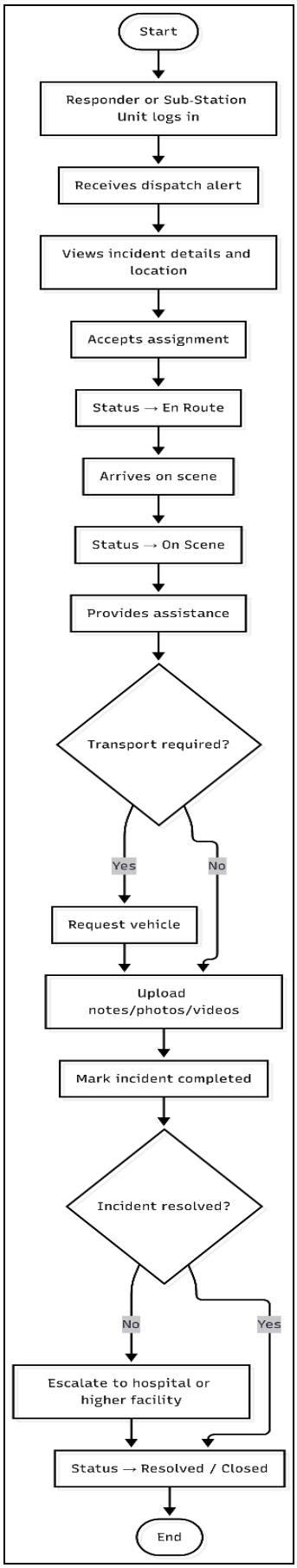
Monitor response progress

- The dispatcher continuously tracks the status and progress of the emergency response until completion.

End

- The workflow ends once the emergency request has been successfully handled and monitored.

Responder Action Flow



Start → Responder or Sub-Station Unit logs in

- The process starts when a responder or unit logs on to the system to ensure authorized access.

Receives dispatch alert

- The responder gets a new incident assignment.

Views incident details and location

- The responder receives the information from the dispatcher such as incident type and location.

Accepts assignment

- The respondent confirms they will handle the case.

Status → En Route

- The system changes the responder's status to "En Route" as they travel to the incident

Arrives on scene

- The responder is on the scene of the incident.

Status → On Scene

- The status is changed to "On Scene" for presence at the incident site.

Provides assistance

- The responder initiates assistance or begins to control the emergency.

Transport required?

- If transport is needed, the responder requests a vehicle.
- If not, they continue assistance without transport.

Upload notes/photos/videos

- The responder uploads notes, photos, or video to document the incident for record-keeping.

Mark incident completed

- The case gets closed by the responder when assistance is provided.

Incident resolved?

- If the incident is not resolved, it is escalated to a hospital or higher facility.
- If resolved, the process continues.

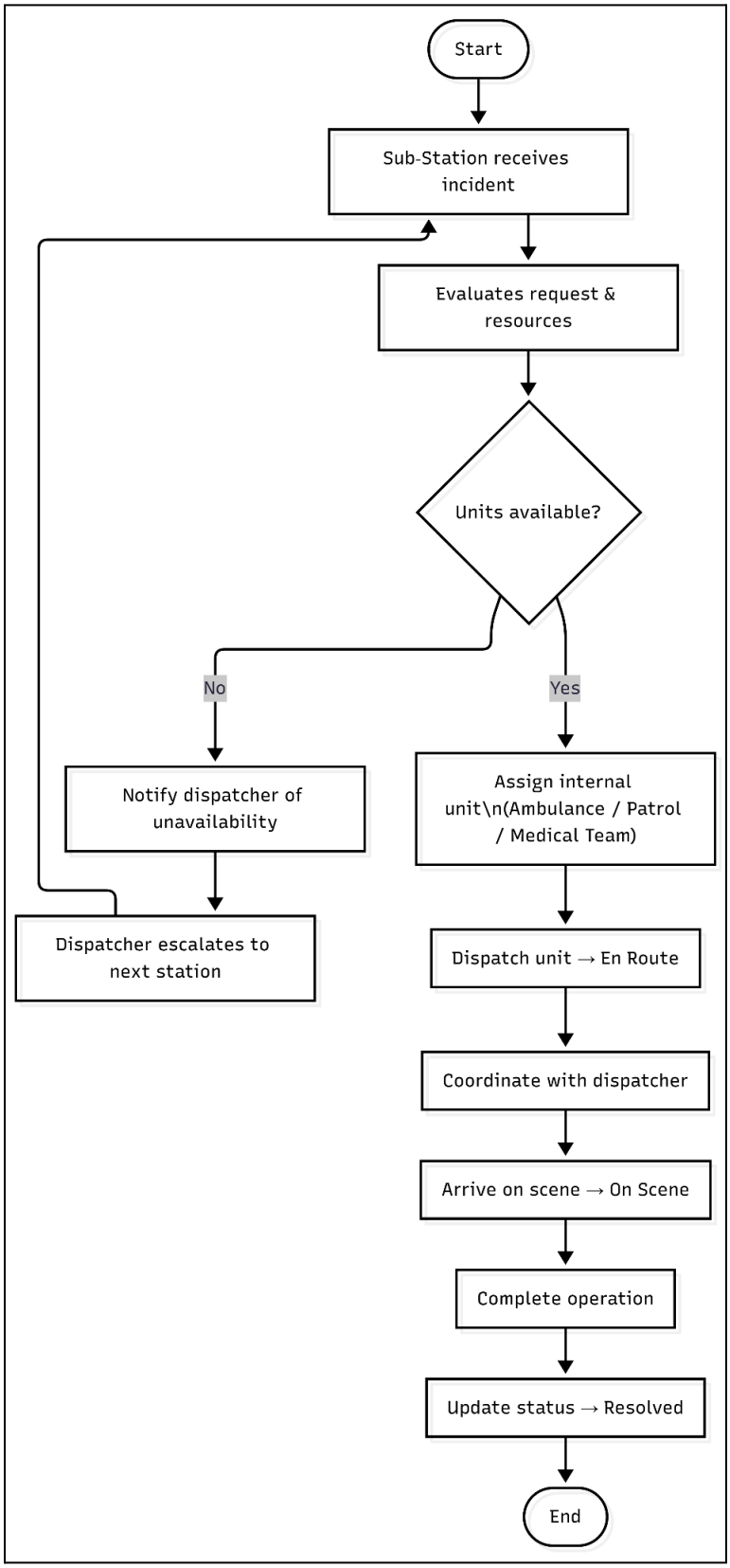
Status → Resolved / Closed

- The incident is Closed and marked as Resolved in the system.

End

- The workflow completes once the incident is officially closed.

Personnel Management Flow



Start → Sub-Station receives incident

- The process begins when a sub-station is notified of a new incident.

Evaluates request & resources

- The sub-station reviews the incident details and checks available resources.

Units available?

If no units are available:

- The sub-station notifies the dispatcher of unavailability.
- The dispatcher escalates the incident to the next nearest station.
- The process loops back to "Sub-Station receives incident."

If units are available:

- The sub-station assigns an internal unit (Ambulance, Patrol, or Medical Team).
- The unit is dispatched.

Dispatch unit → En Route

- The assigned unit begins traveling to the incident location.

Coordinate with dispatcher

- The sub-station maintains communication with the dispatcher to update progress.

Arrive on scene → On Scene

- The unit reaches the incident location and status is updated to "On Scene."

Complete operation

- The unit provides assistance and completes the required emergency response.

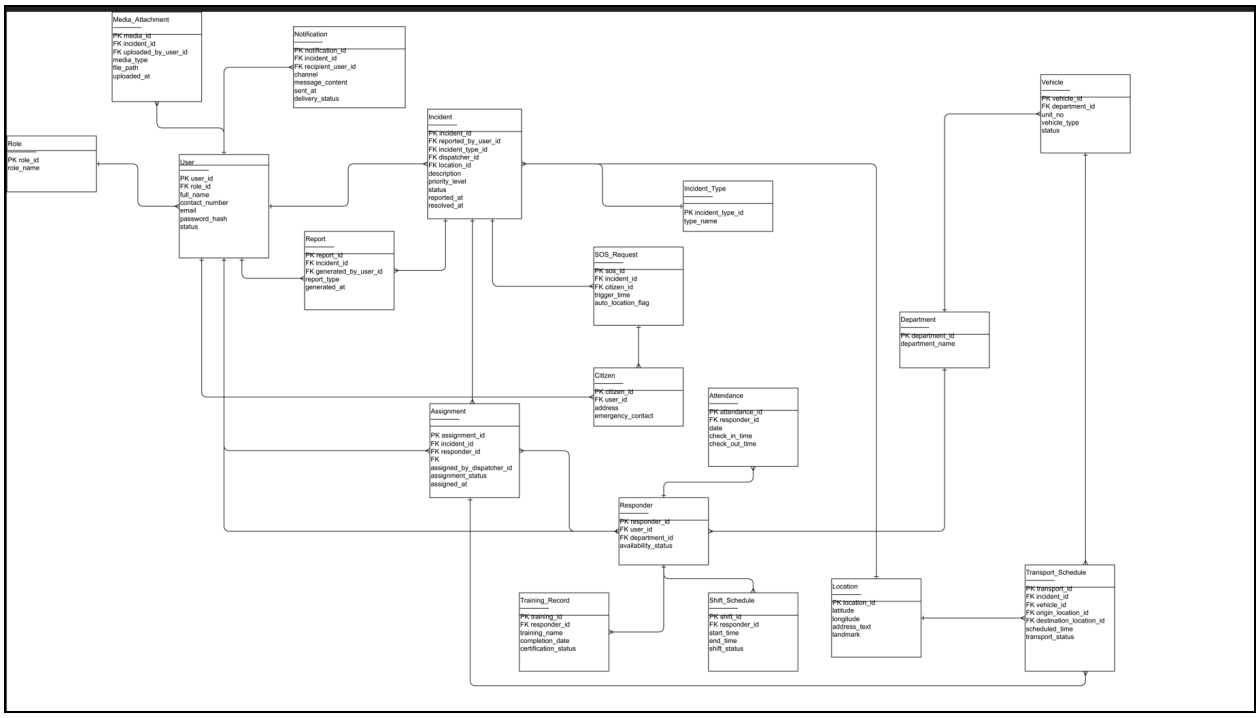
Update status → Resolved

- The incident status is updated to "Resolved" once the operation is finished.

End

- The workflow completes when the incident is officially closed.

2.1.4 Entity Relationship Diagram



Core Tables

Role

- Stores the types of users in the system (e.g. Admin, Dispatcher, Responder, Citizen)
- role_id - unique identifier for each role
- role_name - name of the role

User

- Central table for all system users regardless of their role
- user_id - unique identifier for each user
- role_id - links to Role table to define what type of user they are
- full_name - complete name of the user
- contact_number - phone number
- email - email address used for login and notifications
- password_hash - encrypted password
- status - whether the account is active or inactive

Department - Represents emergency departments (e.g. Police, Fire, Medical))

- department_id - unique identifier
- department_name - name of the department

Responder - Extends User for people who respond to incidents (e.g. police officers, paramedics)

- responder_id - unique identifier
- user_id - links to User table
- department_id - links to Department table to show which department they belong to
- availability_status - whether the responder is available, busy, or off duty

Citizen - Extends User for civilians who can report incidents or send SOS

- citizen_id - unique identifier
- user_id - links to User table
- address - home address of the citizen
- emergency_contact - contact person in case of emergency

Incident Tables

Incident_Type - Lookup table for categories of incidents (e.g. Fire, Flood, Medical Emergency, Crime)

- incident_type_id - unique identifier
- type_name - name of the incident category

Location - Stores geographic information for any location used in the system

- location_id- — unique identifier
- latitude - GPS latitude coordinate
- longitude - GPS longitude coordinate
- address_text - human readable address
- landmark - nearby landmark for easier identification

Incident - The main table that records every emergency incident in the system

- incident_id - unique identifier
- reported_by_user_id - links to User who reported the incident
- incident_type_id - links to Incident_Type to classify the incident
- dispatcher_id - links to User (dispatcher) who is managing the incident
- location_id - links to Location where the incident occurred
- description - detailed description of the incident
- priority_level - urgency level (e.g. Low, Medium, High, Critical)
- status - current state (e.g. Open, In Progress, Resolved)
- reported_at - timestamp when incident was reported
- resolved_at - timestamp when incident was resolved

SOS_Request - Records emergency SOS alerts triggered by citizens

- sos_id - unique identifier
- incident_id - links to the Incident created from this SOS
- citizen_id - links to Citizen who triggered the SOS
- trigger_time - exact time the SOS was triggered
- auto_location_flag - whether the location was automatically detected via GPS

Response Tables

Assignment - Tracks which responders are assigned to which incidents

- assignment_id - unique identifier
- incident_id - links to the Incident being responded to
- responder_id - links to Responder assigned to the incident
- assigned_by_dispatcher_id - links to User (dispatcher) who made the assignment
- assignment_status - current status (e.g. Assigned, En Route, On Scene, Completed)
- assigned_at - timestamp when assignment was made

Vehicle - Stores emergency vehicles belonging to departments

- vehicle_id - unique identifier
- department_id - links to Department that owns the vehicle
- unit_no - vehicle unit number or plate
- vehicle_type - type of vehicle (e.g. Ambulance, Fire Truck, Police Car)
- status - whether the vehicle is available, deployed, or under maintenance

Transport_Schedule - Records vehicle dispatch and transport details for an incident

- transport_id - unique identifier
- incident_id - links to the Incident requiring transport
- vehicle_id - links to Vehicle being dispatched
- origin_location_id - links to Location where vehicle departs from
- destination_location_id - links to Location where vehicle is headed
- scheduled_time - planned departure time
- transport_status - current status (e.g. Scheduled, En Route, Arrived)

Communication Tables

Notification - Records all notifications sent to users about incidents

- notification_id - unique identifier
- incident_id - links to the Incident the notification is about
- recipient_user_id - links to User who receives the notification
- channel - delivery method (e.g. SMS, Email, Push Notification)
- message_content - the actual message sent
- sent_at - timestamp when notification was sent
- delivery_status - whether it was delivered, failed, or pending

Report - Stores generated reports for incidents

- report_id - unique identifier
- incident_id - links to the Incident being reported on
- generated_by_user_id - links to User who generated the report
- report_type - type of report (e.g. Incident Summary, Response Time, After Action)
- generated_at - timestamp when report was created

Media_Attachment - Stores photos, videos, or files uploaded related to an incident

- media_id - unique identifier
- incident_id - links to the Incident the media belongs to
- uploaded_by_user_id - links to User who uploaded the file
- media_type - type of file (e.g. Image, Video, Document)
- file_path - storage path or URL of the file
- uploaded_at - timestamp when file was uploaded

Responder Management Tables

Attendance - Tracks daily check-in and check-out of responders

- attendance_id - unique identifier
- responder_id - links to Responder
- date - the date of attendance
- check_in_time - time the responder checked in
- check_out_time - time the responder checked out

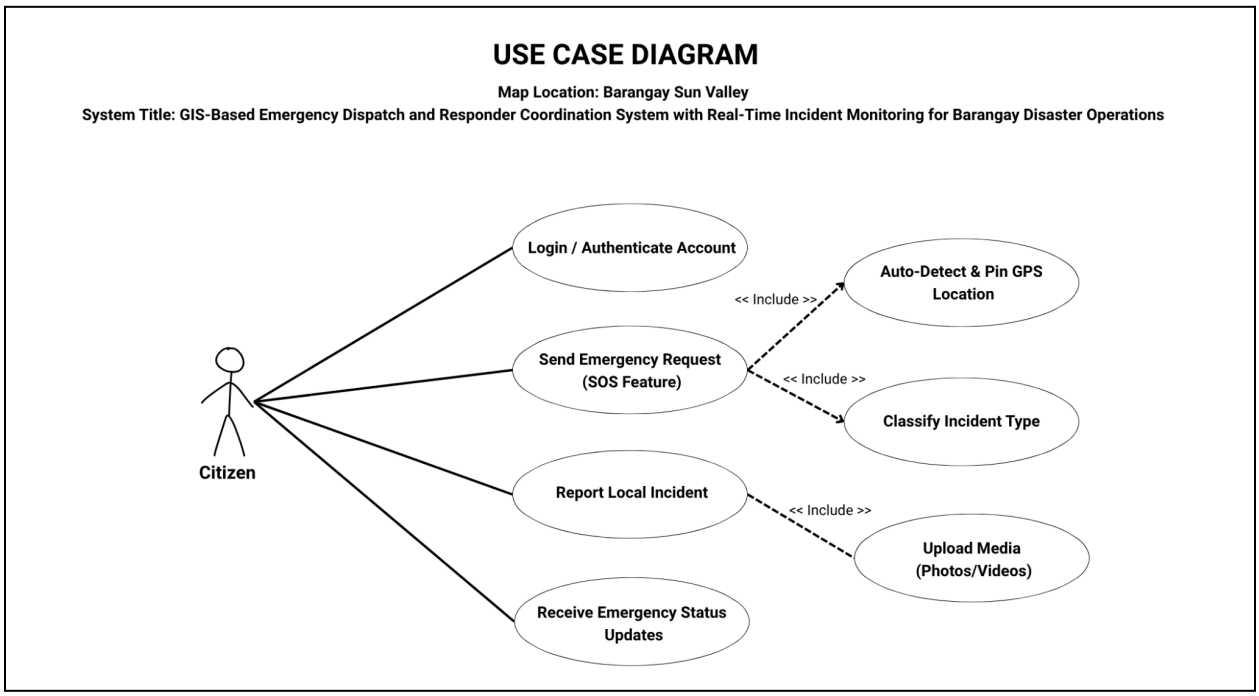
Shift_Schedule - Manages work shifts assigned to responders

- shift_id - unique identifier
- responder_id - links to Responder
- start_time - shift start datetime
- end_time - shift end datetime
- shift_status - current state (e.g. Upcoming, Active, Completed)

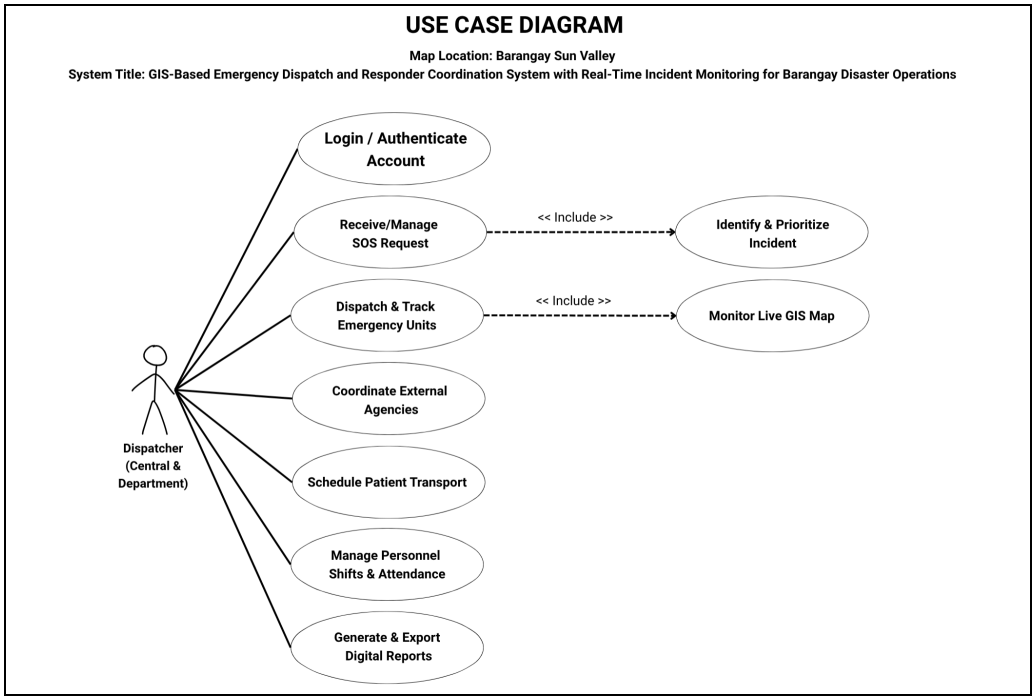
Training_Record - Tracks training and certifications completed by responders

- training_id - unique identifier
- responder_id - links to Responder
- training_name - name of the training program
- completion_date - date training was completed
- certification_status - whether certification was passed, failed, or pending

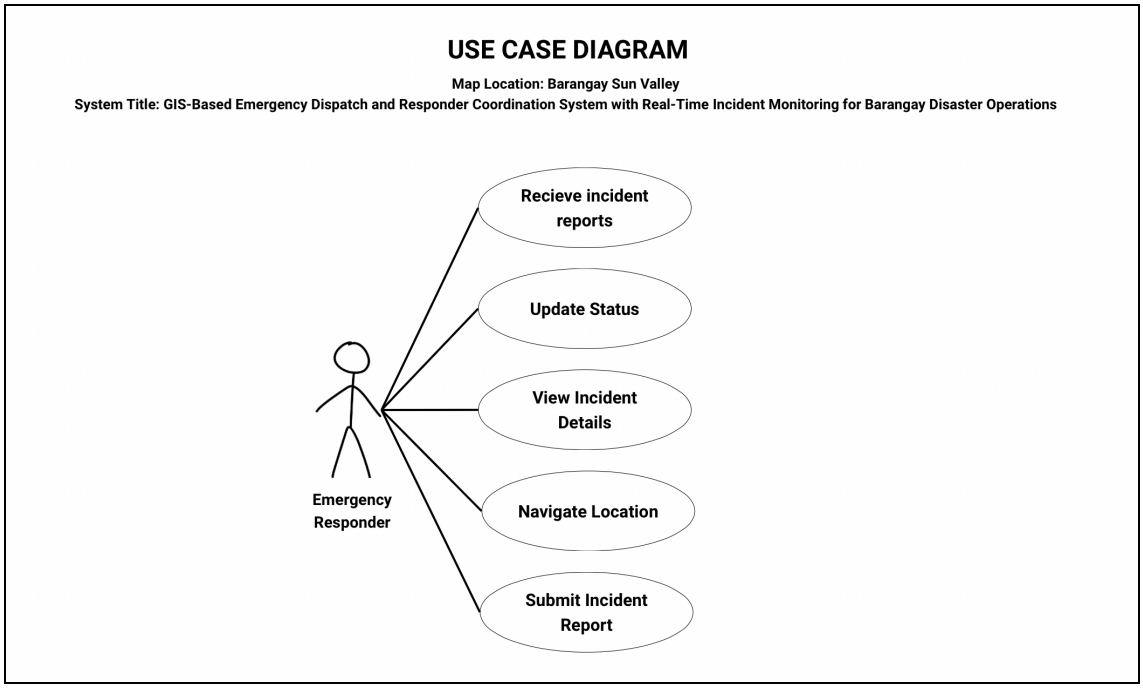
2.1.5 Use Case Diagram



The use case diagram for the GIS-based emergency dispatch and responder coordination system with real-time incident monitoring for barangay disaster operations shows that the citizens of Barangay Sun Valley are the main users and beneficiaries of the system. SOS allows citizens to log in and submit requests during emergencies, locating where a citizen is by using their GPS location when they report an emergency request. It organizes the reports based on the type of issue. Users can also upload photos or other files to report local events and help inform others about what is happening in their area. They can also check updates on their reports and see the progress of ongoing operations. The best part yet was that our system can be disabled instantly and is designed for the residents to use in real emergencies.

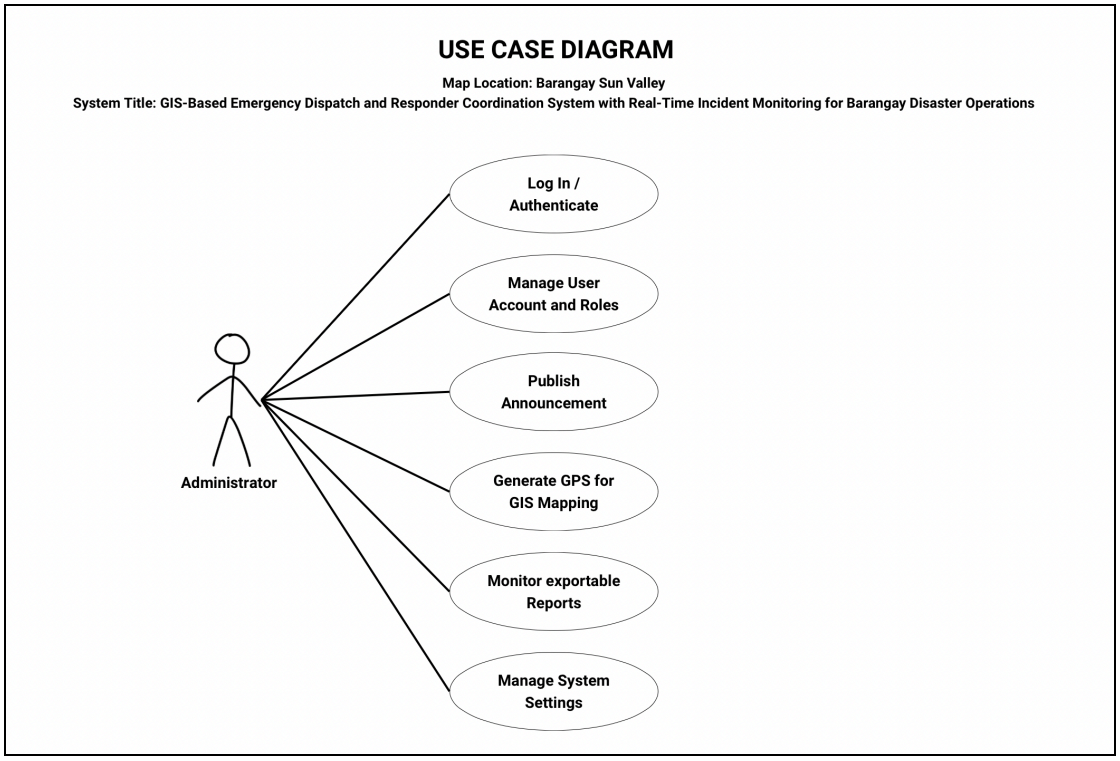


The dispatcher diagram uses a quicker and systematic form of centralizing it in terms of replacing paper-based processes. It also details the responses of barangay dispatchers in Barangay Sun Valley to emergencies and their assistance with some forms of barangay-level response services. For the assurance and security, dispatchers need to log in and verify their accounts before accessing the dashboard. In case a citizen sends out an SOS alert, they are directly routed to the emergency-approved dispatcher for instant help. You take the call as soon as it comes in and are trained to dispatch immediately. This setup eliminates any intermediaries, allowing them to respond rapidly during an emergency. Call and deploy your response teams using Dispatch & Track emergency units. When an emergency is detected, the system allows responders to view their exact location on a real-time GIS map. This will also be used to coordinate agencies from outside and simplify communication with police as well as hospitals and health centers. Dispatchers may also schedule patient transport, monitor equipment status, and handle personnel shifts & attendance. With the generate & export digital reports capability, they can easily save and print response logs and event records.



The emergency responder actor represents the specialized coordination of the platform with the people from the emergency responder such as BEMO, BDRRMO, POPSMO and other volunteer groups like Brigada Onse who handle the active high stress response situation on the mobile devices. The core mobile-operational workflow triggers instantly when a responder is prompted to receive incident alerts, an automated action where incoming dispatch payloads and localized civilian SOS distress signals push straight to their active screen. To address the tracking bottlenecks safely the responder is ready to update their status all throughout the crisis or the emergency. They can choose and let other people see whether they are Enroute, On Scene or Available with just one click.

The actor can view the incident details to assess the situation; they can see the phone number and the specific emergency category. By using the GIS mapping the responder can use the platform to navigate the location easily. The submitted report can be used as a log such as log records, dispatch arrival data, dispatch pull out data and summaries directly the file of the emergency that occurs so that they can share it and have an exportable file.



The System Administrator serves as the backbone of the Barangay Sun Valley Emergency PWA with high-level data-management and administrative privileges not available to the front-line civilian client. The actor starts its work with login and authenticating the session. Here credentials are verified by Firebase architecture and workspace is cleared for the access. Once the administrator has access, they take control of all user accounts and roles. The latter capability allows auditing of registration queues, activation of pending accounts of civilians or responders, as well as configuring users' permissions in a way that no sensitive data would be exposed.

Analytically speaking, this role will involve creating the data needed for GIS mapping. This will include managing active coordinate layers, establishing geographical boundary layers, and performing spatial analysis on incidents. Finally, the actor handles core system settings, maintaining an administrative hold on background application variables, system maintenance activities, and automated configuration overrides to ensure maximum application uptime.

2.2 Requirements Analysis

2.2.1 Functional Requirements

The following shall support the following core functionalities.

Feature	Description	Expected Output
User Authentication	The system allows authorized users to safely access the platform.	Users can log in and access functionality based on given roles.
SOS Emergency Request	The system allows citizens to submit emergency requests with GPS location.	Emergency requests and citizen location are received by the dispatcher.
Incident Management	The system helps dispatchers to monitor and manage emergency occurrences.	Real-time status and updates and incident information are shown.
Responder Dispatching	The system allows dispatchers to assign responders depending on availability and location.	Emergency duties and notifications are distributed to responders.
GIS mapping and GPS tracking	The system allows real-time tracking and mapping of incidents and responses.	The map shows the current locations of events and responses.
Notification system	The system provides SMS and in-app alerts during emergencies.	Emergency notifications are sent to departments and responders right away.
Status update monitoring	Responders can update their operational status through the system.	The dashboard displays updated responder statuses.
Digital Documentation	The system maintains incident reports, records, and uploaded information digitally.	The database stores and retrieves incident records.
Personnel Management	The system manages personnel attendance, schedule, and training records.	Schedules and personnel data are updated and monitored.
Report Generation	The system generates exportable emergency and incident reports.	Reports are available for download in CSV, Excel, and PDF formats.

2.2.2 Non-functional Requirements

Feature	Description	Criteria
Usability	The system will provide an intuitive and user-friendly interface for citizens, dispatchers during emergency situations.	Users should be able to navigate and use the system with minimum training and confusion.
Performance	Notification, GPS updates, and emergency requests should all be handled quickly and effectively by the system.	The system should respond to user activities and provide updates within a few seconds under normal working conditions.
Security	The system must protect unauthorized access to user accounts, incident data, and emergency records.	The system should incorporate secure login authentication, role-based access control, and protected data storage.
Availability	The system should continue to function and be accessible when emergency services are required.	The system should be accessible 24/7 with minimal downtime during operations.
Scalability	Future improvement in the terms of departments, users, and new features should be supported by the system.	The system should continue to function successfully as the number of users and stored data increases.

2.3 Requirements Documentation

2.3.1 Software Requirements Specification

The following section identifies the software tools, frameworks, platforms, and technologies required for the development, deployment, and maintenance of the proposed system.

Development Tools and Frameworks:

a) Frontend Development

- React.js
- HTML5
- CSS3
- JavaScript

- Bootstrap 5 / Tailwind CSS
- Mapbox API
- Progressive Web Application (PWA) Technology

b) Backend Development

- Node.js
- Express.js
- MongoDB
- Mongoose
- Socket.io
- RESTful API
- Firebase Cloud Messaging (FCM)

c) Security and Authentication

- JSON Web Token (JWT)
- Role-Based Access Control (RBAC)
- bcrypt.js (Password Encryption)
- HTTPS / SSL Encryption
- Express Validator
- CORS Middleware
- Helmet.js

2.3.2 Hardware Requirements Specification

These are the minimum and recommended hardware requirements for system development, testing, deployment and operation.

Development Environment:

Developer Machines (for coding and testing)

a) Processor

- Minimum: Intel Core i3 (10th Generation) / AMD Ryzen 3
- Recommended: Intel Core i5 (11th Generation or higher) / AMD Ryzen 5

b) Memory (RAM)

- Minimum: 8GB RAM

- Recommended: 16GB RAM

c) Storage

- Minimum: 256GB SSD
- Recommended: 512GB SSD or higher

d) Graphics

- Minimum: Integrated Graphics (Intel UHD / AMD Vega)
- Recommended: Dedicated GPU (NVIDIA GTX / AMD Radeon)

e) Operating System

- Minimum: Windows 10
- Recommended: Windows 11

2.3.3 Testing Devices

- Computer(s) for dashboard or web testing (Desktops or Laptop)
- The Android smartphones are ideal for PWA testing, GPS tracking, and notifications.
- Mobile device with GPS to achieve real-time validation of location

2.3.4 Network Requirements

- A reliable Wi-Fi or cell phone signal for immediate access to messages
- GPS tracking, notifications and cloud syncing require Internet access.

2.3.5 Constraints and Assumptions

Constraints:

- Needs a stable internet connection to perform real-time GPS tracking, alerts and data synchronization.
- The accuracy of GPS may change based on the capabilities of the mobile device, the strength of the signal and the surrounding environment.
- Availability of third party services and reliability of networks could be a factor in the delivery of SMS and notifications.
- The primary target audience are web browsers and Android-based mobile devices.
- In emergency operations performed simultaneously with high user activity, limited hardware resources may affect the performance of the system.

Assumptions:

- Users are able to use mobile devices and Web applications in a basic way.
- GPS-enabled and internet connected devices are given to responders and personnel.
- The involved organizations will be exchanging correct and up-to-date emergency response information.
- User accounts, information about the respondents and incident records will be handled and maintained appropriately by authorized staff.

2.4 Development and Pre-Testing procedures

2.4.1 Development

- Build Progressive Web Application (PWA) using Visual Studio.
- Firebase-powered backend with authentication, real-time DB, and secure transactions.
- Implement GIS mapping and GPS tracking by responders.
- Enable role-based access (citizens, dispatchers, responders, admins).
- Create notification system (SMS/App alerts).

2.4.2 Pre-Testing

- Developmental and faculty testing of internal.
- Test citizens and admins login/authentication.
- Check the feature SOS and auto-location pinning.
- Check GPS accuracy and real time tracking.
- Test notification system (alerts, announcements).
- Validate documentation/report generation.

2.4.3 Operation Procedures

a) **For Citizens (Users):**
    - Download and install the mobile app.
    - Use a valid email address to register.
    - For logout, you must use the same verified credentials used to login.
    - SOS messages with Auto Location.

- Get live alerts and notifications.

b) For Administrators/Dispatchers:

- Use verified administrator credentials to log in.
- Add, edit, and delete responder accounts.
- Establish an emergency/voting time (beginning and end).
- Make announcements (deadlines, reminders).
- Get live report of incident submissions through Firebase backend.
- Complete and release official reports/results.

2.4.3 Testing Procedures

I. Function-Based Testing

Test Case	Steps to be Taken	Expected Output	Result
Citizen & Personnel Registration	Citizen or responder creates an account → Admin verifies the registration request.	Account activated and secure role assignment applied to database profile.	Passed/Failed
SOS Request Trigger	Citizen sends an SOS → Dispatcher receives automated data payload.	Incident instantly logged with automated real-time GPS coordinates pinned on map.	Passed/Failed
GPS Tracking & Status Updates	Field Responder updates operational status via mobile client platform.	Real-time unit location and updated status layer instantly visible on central dispatcher map.	Passed/Failed

Test Case	Steps to be Taken	Expected Output	Result
Emergency Announcement	Admin publishes a critical safety announcement → Citizen receives alert.	System broadcast push notification delivered to active devices in real time.	Passed/Failed
System Authentication	User attempts login with unverified, deactivated, or incorrect credentials.	Access explicitly denied with a descriptive validation error message displayed.	Passed/Failed
Report Generation	Admin exports historical aggregate incident response data logs.	Comprehensive PDF/Excel/CSV generated correctly with verified metrics and complete timestamps.	Passed/Failed

II. Function-Based Testing

Test Module	Test Scenario	Result
Navigation & Access Matrix	Can users successfully access all pages allowed by their assigned role (Dashboard, SOS Tracking, Responders Monitor, Analytical Reports)?	Passed/Failed
Announcement Push	Does the end-user device receive instantaneous notification streams the moment a new announcement is broadcasted by the admin?	Passed/Failed

Real-Time Alerts Engine	Are incoming emergency dispatch vectors and assignment alerts delivered instantly to the target responders' mobile client interfaces?	Passed/Failed
SOS Data Integrity & Precision	Is the transmission of the SOS payload (coordinates, time, user identity) correctly saved, buffered offline if needed, and accurately reflected in the backend database?	Passed/Failed
Authentication Guard	Can only verified citizens, validated field responders, authorized dispatchers, and systems administrators bypass security checkpoints to access workspaces?	Passed/Failed
Incident Finality Assurance	Once an active crisis lifecycle is officially marked as "Closed" by the central dispatcher, is the historical record locked securely from being altered or edited again?	Passed/Failed
Admin Operational Controls	Can the administrative user successfully activate/deactivate emergency broadcast periods and dynamically manage individual vehicle/responder visibility states?	Passed/Failed

2.5 Implementation Plan (Infrastructure/ Deployment)

2.5.1 Infrastructure

- **Development Environment:**
  - Visual Studio as the main IDE for Progressive Web and Mobile Apps.
  - Using GitHub/GitLab for version control, allowing collaborative development and rollback.
- **Backend Services:**
  - Firebase Authentication for secure role-based access (citizens, dispatchers, responders, admins).

    - FireBase Realtime Database for incident notes, responder availability and scheduling transports.
    - Cloud Firestore for structured reporting and archival.
- **GIS & GPS Integration:**
    - Google Maps API for real-time incident visualization.
    - GPS tracking modules on responder devices for realtime tracking.
- **Security & Compliance:**
    - Encryption of all communications (SSL/TLS).
    - Role-based access control to prevent unauthorized use.
    - Barangay disaster protocols with audit logs for accountability and compliance.
- **Hardware & Connectivity:**
    - Citizens' and responders' mobile devices.
    - A stable internet and back-up power supply at the dispatcher work stations.
    - Cloud-hosting with redundancy to reduce downtime.

2.5.2 Deployment

- **Stage 1: Internal Testing**
    - Conducted by developers and faculty.
    - Focusing on keeping it functional (login, SOS, GPS tracking, notifications).
    - Identify bugs and refine user interface.
- **Stage 2: Beta Testing**
    - Controlled rollout to a small group of students and barangay responders.
    - Collect usability feedback (ease of navigation, clarity of alerts).
    - Stress-test system under simulated multiple incidents.
- **Stage 3: Full Deployment**
    - Official launch during barangay disaster operations.
    - Real-time monitoring by election officers and emergency managers.
    - Incident drills conducted to validate readiness.
- **Risk Mitigation:**
    - Backup servers to ensure continuity during outages.
    - Failover protocols for database and notification services.

- Contingency plan for manual dispatch if system downtime occurs.

2.5.3 Support & Sustainability

- **Continuous Updates:**
  - Regular patches for bug fixes and performance improvements.
  - Feature enhancements based on feedback (e.g., multimedia SOS reporting).
- **Technical Support:**
  - Dedicated support team available during election/emergency periods.
  - Hotline and chat support for citizens and responders.
- **Training & Capacity Building:**
  - Workshops for dispatchers and responders on system use.
  - Citizen friendly user guides and quick-start guides.
- **Monitoring & Evaluation:**
  - Performance Reporting monthly (Response Times, Incident Resolution).
  - Review of effectiveness with barangay officials annually.
- **Scalability:**
  - Future integration to city level or national emergency systems.
  - Expansion to include predictive analytics (AI-based incident forecasting).
- **Sustainability:**
  - Funding for maintenance and upgrading systems.
  - Ongoing research and development with local universities.

Phase	**Activities**	**Details / Explanation**	**Academic Timeline**
Phase 1: Initial Research & Design	**Research and Documentation**	Conduct background research, interviews with personnel and literature review. Write Chapter I–II of the thesis (Introduction & Methodology). Prepare diagrams, system design, and	3rd Year –Second Semester (Current)

Phase	Activities	Details / Explanation	Academic Timeline
		initial documentation.	
	Planning and Design	Create system architecture diagrams, define modules (dispatch, SOS, GIS, reporting), and finalize scope and delimitations. Prepare mockups and initial prototypes for faculty review.	3rd Year –Second Semester (Current)
Phase 2: Prototype Architecture	**Prototype Development**	Begin coding small modules such as login, SOS button, and basic GIS map integration. Conduct internal testing with classmates and faculty.	4th Year – First Semester
	Refinement	Improve user interface, add role-based access, and prepare for larger testing.	4th Year – First Semester
Phase 3: Core Implementation	**Full System Development**	Focus on building the complete Progressive Web Application (PWA). Integrate Firebase authentication, real-time database, GPS tracking, and reporting modules.	4th Year – First Semester
	Stage 1: Internal Testing	Developers and faculty test login, SOS, GPS tracking,	4th Year – First Semester

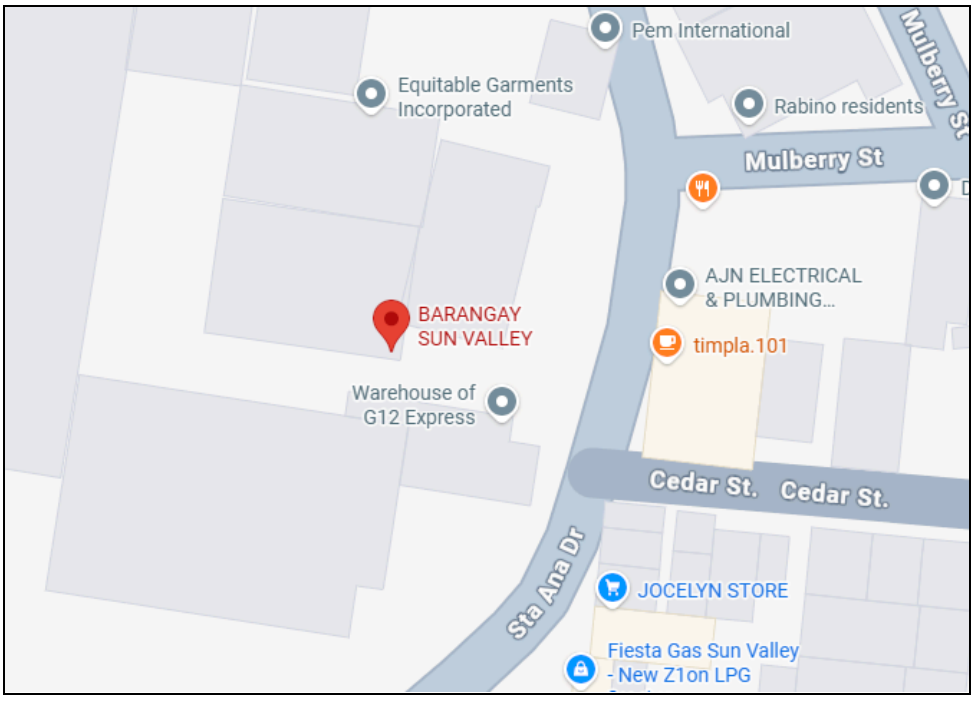
Phase	Activities	Details / Explanation	Academic Timeline
		notifications. Identify bugs and refine user interface.	
	Stage 2: Beta Testing	Controlled rollout to a small group of students and barangay responders. Collect usability feedback and stress-test with multiple incidents.	Mid 4th Year – First Semester
	Stage 3: Full Deployment	Official launch during barangay disaster drills or operations. Real-time monitoring by barangay officials and responders. Conduct incident drills to validate readiness.	End of 4th Year – First Semester
Phase 4: Optimization & Roadmaps	**Support & Sustainability**	Conduct training workshops for dispatchers and responders. Provide manuals and guides for citizens. Monitor system performance monthly and prepare final thesis defense.	End of 4th Year – First Semester
	Scalability and Future Plans	Explore integration with city-level systems and predictive analytics. Partner with local universities for continuous research and development.	Long-term (Beyond Capstone)

2.6 Statistical Treatment of data

2.6.1 Data Gathering Technique

The researchers will conduct semi-structured interviews with personnel from the Barangay Emergency Management Office (BEMO), Sun Valley Barangay Disaster Risk Reduction Management Office (BDRRMO), and Brigada Onse Sun Valley Fire and Rescue Volunteer Group, Incorporated to gather data. These interviews seek to identify the issues, challenges, workflow and system needs for emergency response, communication, incident monitoring and documentation in Barangay Sun Valley. The data collected will be used to support development of the proposed GIS based Emergency Dispatch and Responder Coordination System. Once the system is developed, the team will test and evaluate it with the involved personnel to check its functionality, usability and efficiency.

2.6.2 Setting of the Study



This study will be conducted in Barangay Sun Valley, Parañaque City. The Barangay Emergency Management Office, the Sun Valley Barangay Disaster Risk Reduction Management Office and the Brigada Onse Sun Valley Fire and Rescue Volunteer Group, Incorporated shall be covered. The setting is vital to the study as these organizations do emergency response operations in the barangay. The site will allow researchers to observe real-time emergency response operations and collect data to evaluate the effectiveness of the proposed GIS-based Emergency Dispatch and Responder Coordination System in enhancing coordination, location tracking and documentation.

2.6.3 Subject of the Study

Individuals working at the Barangay Emergency Management Office (BEMO), Sun Valley Barangay Disaster Risk Reduction Management Office (BDRRMO), and Brigada Onse Sun Valley Fire and Rescue Volunteer Group, Incorporated in Barangay Sun Valley, Paranaque City will serve as the subjects for this study. Utilizing total population sampling, all 44 personnel are included in the data collection. This group consists of 19 BEMO personnel and 25 POPSMO personnel. With all of them involved the researchers will have comprehensive data regarding emergency response operations, coordination, and incident documentation.

2.6.4 Sampling Technique

The study will use total population sampling which includes all 44 personnel of Barangay Emergency Management Office (BEMO), Sun Valley Barangay Disaster Risk Reduction Management Office (BDRRMO) and Brigada Onse Sun Valley Fire and Rescue, Volunteer Group, Incorporated as respondents. This approach is feasible as the size of the population is small and all personnel are involved directly in emergency response operations. Their participation is necessary for the study to have full and reliable data.

2.6.5 Statistical Treatment of Data

The statistical tools to be utilized by the researchers in analyzing and interpreting the data gathered for the study are Frequency Count and Weighted Mean. The statistical treatments will help in evaluating the proposed GIS-based Emergency Dispatch and Responder Coordination System, and in summarizing the responses of the participants involved in the study.

Frequency Count

Frequency Count will be used to describe and present the profile of the respondents (e.g. role, department, years of service) and the distribution of their responses during the system evaluation. It is a statistical tool which measures how often a particular response or category appears in the data collected.

Where:

- f = frequency of responses

Weighted Mean

The Weighted Mean based on the ISO 25010 software quality characteristics; usability, performance efficiency, security, availability and scalability will analyze and interpret the respondents' evaluation of the proposed system. The respondents' answers will be measured on a five point Likert scale.

The formula for Weighted Mean is:

$$\bar{X} = \frac{\sum(f \cdot x)}{N}$$

Where:

- x = weighted mean
- $\sum fx$= sum of the product of frequency and weight
- n = total number of responses

The following Likert Scale will be used for interpreting the responses:

Likert-Scale	**Likert-Scale Description**
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree